Market Reactions to Natural Disasters: An Event Study of Cyclone Ditwah's Impact on the Colombo Stock Exchange

Abstract

Natural disasters inflict severe macroeconomic shocks that ripple through financial markets, yet the dynamics of such shocks in emerging-economy exchanges remain understudied. Cyclone Ditwah made landfall in eastern Sri Lanka on 28 November 2025, causing an estimated USD 4.1 billion in direct damage — approximately 4% of GDP — and triggering the Colombo Stock Exchange's (CSE) worst weekly decline since November 2022. This project employs a classical event-study framework on a self-collected dataset of daily Open-High-Low-Close (OHLC) and volume data for 296 CSE-listed stocks, spanning February 2025 to February 2026. We aim to quantify abnormal returns and cumulative abnormal returns across the full market and disaggregated by sector, characterise post-disaster volatility and liquidity dynamics.

Keywords

event study, natural disaster, stock market, Colombo Stock Exchange, Cyclone Ditwah, emerging markets, abnormal returns, disaster economics

1 Problem Motivation

Sri Lanka's economy, still recovering under an IMF Extended Fund Facility arrangement following the 2022 sovereign debt crisis, suffered a major setback when Cyclone Ditwah struck the island on 28 November 2025. The World Bank estimated direct physical damage at USD 4.1 billion [1], affecting agriculture (58,340 hectares of paddy inundated), critical infrastructure (247 km of roads and 35 bridges damaged), and displacing over 209,000 people across all 25 districts [2].

Financial markets responded sharply. The CSE All-Share Price Index (ASPI) fell 3.04% on the first trading session after landfall, extending to a weekly loss of 5.35% — the worst since mid-November 2022 [4]. Net foreign outflows from the exchange accelerated during December 2025.

Despite these observable effects, no peer-reviewed study has systematically quantified the impact of Cyclone Ditwah on individual stocks or market sectors at the CSE. This gap matters for three stakeholder groups: investors who need to price disaster risk into portfolios; policymakers who must design market stabilisation interventions; and researchers who require empirical evidence on emerging-market resilience.

Our project addresses this gap by combining a purpose-built, high-resolution CSE dataset with rigorous event-study and time-series methodology to produce a quantitative analysis of Cyclone Ditwah's capital-market impact.

2 Identified Data Sources

2.1 Primary Dataset — CSE Historical Price Data

We collected daily trading data for 296 stocks listed on the Colombo Stock Exchange via web scraping. The dataset was assembled using a custom Python pipeline querying daily stock data, covering approximately one trading year.

Table 1: Primary dataset characteristics

Attribute	Value
Coverage	296 CSE-listed stocks
Date range	23 Feb 2025 – 19 Feb 2026
Total records	66,958 rows
Fields	Symbol, date, high, low, close, volume
Method	Webscraping public data
License	Public / open access

2.2 Supplementary Sources

The disaster timeline was sourced from WHO and OCHA Situation Reports [2, 3] to obtain precise event dates including landfall, peak flooding, and the declaration of emergency. Sector classification was derived from the CSE sector listings, assigning each stock to an industry group. Macroeconomic control variables — exchange rate, interest rates, and inflation — were obtained from Central Bank of Sri Lanka monthly statistics and the IMF Country Report [5]. CSE ASPI and S&P SL20 index data serve as benchmark indices for computing expected returns.

3 Proposed Methodology

3.1 Event Study Framework

We adopt the standard event-study methodology [6]. The event date ($t = 0$) is 28 November 2025 (landfall), with an event window of $[-5, +30]$ trading days and an estimation window of $[-120, -6]$ trading days.

The market model relates individual stock returns to market returns:

$$R_{it} = \alpha_i + \beta_i R_{mt} + \varepsilon_{it} \tag{1}$$

Abnormal returns are computed as the residual from the fitted model:

$$AR_{it} = R_{it} - \left(\hat{\alpha}_i + \hat{\beta}_i R_{mt}\right) \tag{2}$$

Cumulative abnormal returns aggregate over the event window:

$$CAR_i(t_1, t_2) = \sum_{t=t_1}^{t_2} AR_{it} \tag{3}$$

Statistical significance is assessed using cross-sectional t-tests and Wilcoxon signed-rank tests on mean CARs for the full market and each sector sub-group.

3.2 Sector-Level Analysis

After assigning each stock to its CSE sector, we compare mean CARs across sectors. We hypothesise that agriculture and tourism stocks will exhibit significantly larger negative CARs than banking or telecommunications stocks, consistent with the sectoral damage profile of the cyclone.

3.3 Volatility and Liquidity Analysis

We use the daily High–Low spread as a model-free volatility proxy and daily traded volume as a liquidity measure. Paired t-tests compare pre- and post-event averages for both metrics across the full sample and sector sub-groups.

3.4 Recovery Trajectory (Optional)

As an optional extension, we will apply ARIMA and LSTM models to the post-event price series to estimate time to recovery.

3.5 Reproducibility

All data collection, preprocessing, analysis, and visualisation code will be maintained in a public GitHub repository with full documentation and step-by-step execution instructions.

4 Expected Outputs

This project will deliver quantitative estimates of abnormal returns at both the aggregate market level and disaggregated by CSE sectors, providing a rigorous measurement of Cyclone Ditwah's financial impact across the full equity listing. By computing and comparing cumulative abnormal returns across the agriculture, tourism, banking, and construction sectors, the study will identify which sectors proved most and least resilient to the shock — evidence directly relevant to portfolio managers and disaster-risk analysts operating in frontier and emerging markets.

The post-event analysis will characterize the volatility clustering and liquidity contraction that followed the cyclone, using the High–Low spread and traded volume to document how market conditions deteriorated and subsequently normalized.

Finally, the project will release a fully reproducible open-source dataset and codebase: raw OHLC and volume data for used 296 listed stocks, the Python collection pipeline, and end-to-end analysis scripts covering all steps from data ingestion through statistical testing to visualization. This contribution is intended to lower the barrier to future research on emerging South Asian equity markets and on the financial economics of natural disasters more broadly.

References

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[5] International Monetary Fund. IMF Country Report No. 25/339: Sri Lanka. 2025.

[6] A. C. MacKinlay. Event studies in economics and finance. *Journal of Economic Literature*, 35(1):13–39, 1997.